Discussion 4

Problem 1. The probability mass function of a random variable is given by

$$p_X(i) = \frac{c\lambda^i}{i!}; i = 0, 1, 2, \cdots,$$

where λ is a positive value.
(a) Find $P(X = 0)$
(b) Find $P(X > 2)$

Problem 2. An unfair coin for which $P(H) = p$, where $0 < p < 1$. We toss the coin repeatedly until we observe a head for the first time. Let Y be the total number of coin tosses. Find the distribution of Y.

(1) Coin tosses are independent.

(2) In each toss $P(H) = p$ and $P(T) = 1 - p$

(3) Y = the total number of coin tosses until first head

Problem 3. Assume that CDF of the random variable X is given as follows.

$$F(x) = \begin{cases} 0 & x < 0 \\ .06 & 0 \le x < 1 \\ .19 & 1 \le x < 2 \\ .39 & 2 \le x < 3 \\ .67 & 3 \le x < 4 \\ .92 & 4 \le x < 5 \\ .97 & 5 \le x < 6 \\ 1 & 6 \le x \end{cases}$$

(a) Find pmf of X.
(b) Using CDF find $P_X(4)$
(c) $P(1 \le X \le 4)$

Problem 4. A 5-card poker hand is said to be a full house if it consists of 3 cards of the same denomination and 2 other cards of the same denomination (of course, different from the first denomination). Thus, a full house is three of a kind plus a pair. What is the probability that one is dealt a full house?

Problem 5. In an experiment n independent trials is performed. Each trial results in a success with probability p and a failure with probability $1-p$. What is the probability that

(a) at least 1 success

(b) exactly k successes

Write down all the formulas, reasoning and derivation.

Problem 6. It is estimated that 50% of a certain population has a particular illness. A medical test has been developed to detect the illness. The test correctly identifies 98% of those who have the illness, and the probability of a false positive (a healthy person identified as having the illness) is 6%.

Now, if a person tests positive, what is the probability that they are actually healthy?

Problem 7. How many ways are there to distribute 6 different snacks and 9 different toys between 3 kids

(a) without any restrictions;

(b) If the first kid needs to get exactly two snacks and 3 toys;

(c) If all kids need to get exactly two snacks and 3 toys.

Problem 8.

(a) How many ways are there to divide 52 cards between four people so that each person gets 13 cards.

(b) Find the probability that players 1 and 2 have two queen each.

(c) Find the probability that all cards of player 1 are either spades or clubs and he has exactly one king.

Problem 9. There are two identical boxes. The first box contains 6 balls numbered 1 to 6. The second box contains 10 balls numbered 1 to 10. One box is chosen at random and then 4 balls are selected from that box without replacement. Let A be the event that the first box is chosen, B be the event that the second box is chosen and M be the event that the maximum number of the balls chosen is 6. Compute

(a) $P(M|A)$ and $P(M|B)$;

(b) $P(M)$;

(c) $P(A|M)$.

Problem 10. Two cards are randomly selected from an ordinary deck of 52 cards. Let B be the event that both cards are aces. Let E be the event that ace of spades is chosen and F be the event at least one aces is chosen. Find

(a) $P(B|E)$

(b) $P(B|F)$

Problem 11. A box contains 5 blue balls, 6 red balls, and 7 black balls. Two balls are randomly drawn without replacement.

(a) What is the probability they are both blue?

(b) What is the probability they have different colors?

Problem 12. In a certain community, 36 percent of the families own a dog and 22 percent of the families that own a dog also own a cat. In addition, 30 percent of the families own a cat. What is

(a) the probability that a randomly selected family owns both a dog and a cat?

(b) the conditional probability that a randomly selected family owns a dog given that it owns a cat?

Problem 13. Let A, B and C be mutually independent and $P(A) = 1/2$, $P(B) = 1/3$, $P(C) = 1/4$. Let D be the event that exactly one of events A, B and C occurs.

(a) Compute $P(D)$.

(b) Compute $P(A|D), P(B|D), P(C|D)$.

(c) Compute $P(A|(B \cup C))$.

Problem 14. Random variable T is distributed with the following pdf:

$$f(t) = \begin{cases} sin(t) & 0 < t < k \\ 0 & \text{otherwise} \end{cases}$$

(a) What is the value of the constant k?

(b) What is the corresponding CDF?

Problem 15. The probability mass function of X is given in the following table

x	0	1	2	3
$P_X(x)$	0.2	0.4	0.1	0.3

Let $Y = (X - 1)^2$

(a) Compute the probability mass function of Y.

(b) Compute the cumulative distribution function of Y.